

Parent Guide



@twinklparents

We're excited to share this activity with you. If you are interested in finding more engaging, fun and interesting activities for you and your children, then check out these links to different areas of the [Twinkl Parents](#) website.

games



crafts



puzzles



experiments



word searches



What is this resource and how do I use it?

Support your year 6 child to develop their understanding of 3D shapes with this fun Properties of 3D Shapes activity, which includes a variety of tasks for your child to complete as well as a quick guide. All you need to do is download, print off and get learning!

What skills does this practise?

Properties of Shapes

Geometry

3D Shapes

Further Activity Ideas and Suggestions

Have a read of this helpful [parent guide to year 6 maths](#) and check out our useful [Top Tips for Year 6 SATs](#). Take the learning outdoors with this fun [Outdoor 3D Shape Hunt](#) or have a go at this [3D Shapes and Nets Matching Game](#).

Parents Blog



Twinkl Kids' TV



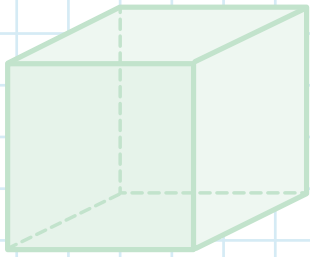
Homework Help



twinkl

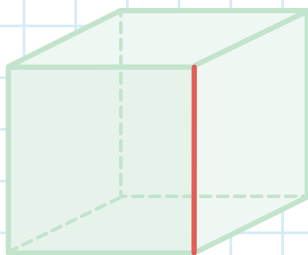
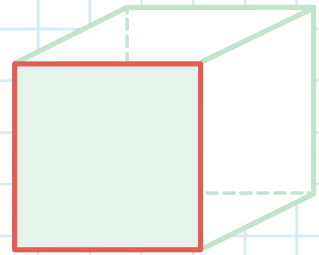
Parents Hub

Properties of 3D Shapes

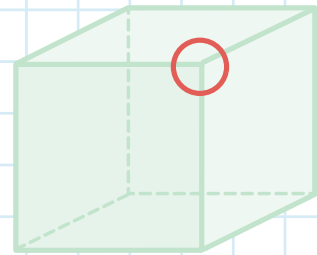


This 3D shape is a **cube**.

The 2D shape shown is one of the cube's **faces**. A cube has **6 faces** in total. They are all **squares**.



Where two faces meet is called an **edge**.
A cube has **12 edges** in total.

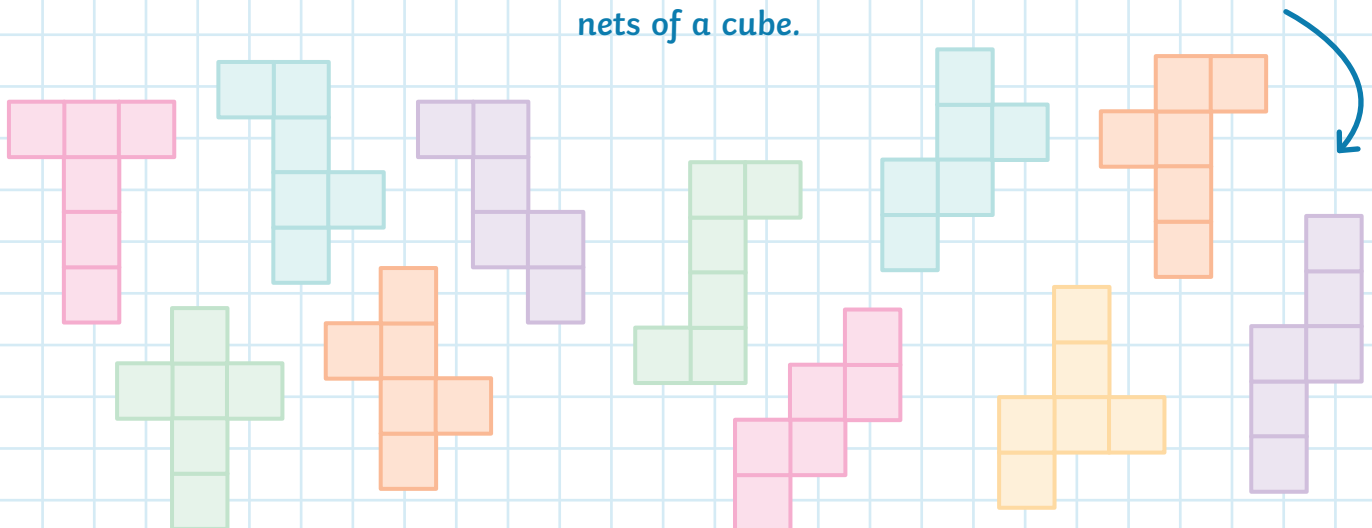


Where the edges meet is called a **vertex**.
A cube has **8 vertices** in total.

3D shapes can be represented as **2D nets**.

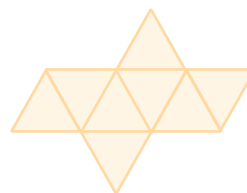
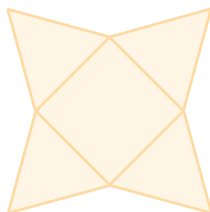
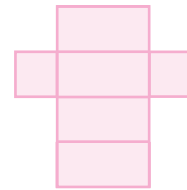
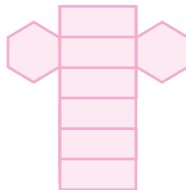
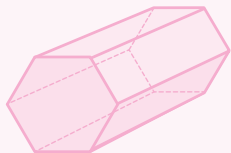
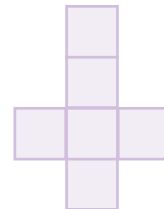
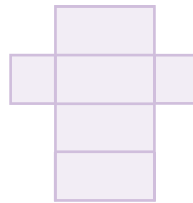
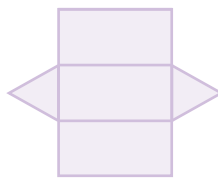
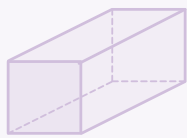
Nets are useful for helping you to see which different 2D shapes make up a 3D shape.

A 3D shape can have **more than one** possible net. For example, there are 11 possible nets of a cube.



3D Shape Nets

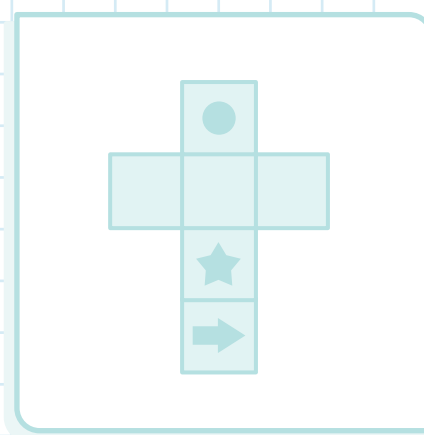
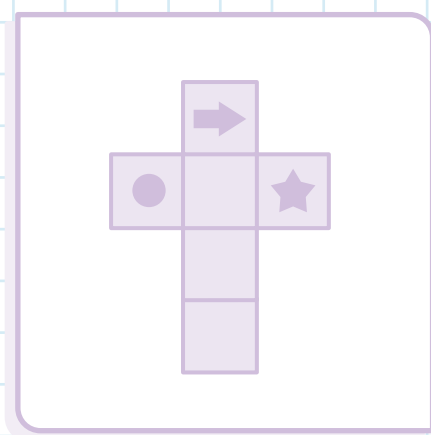
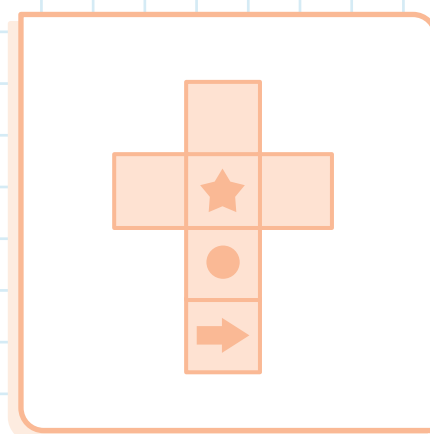
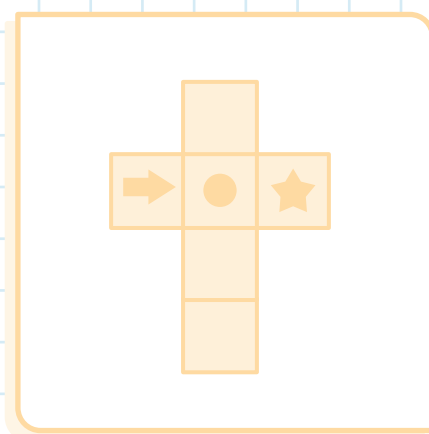
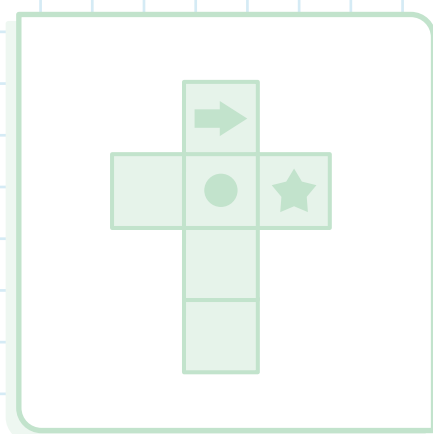
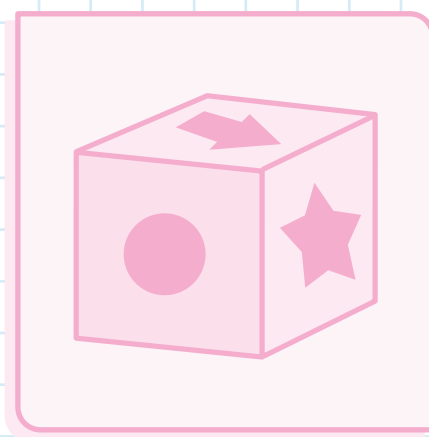
Circle the correct net for each shape.



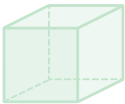
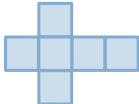
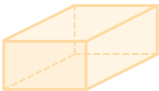
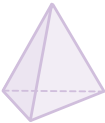
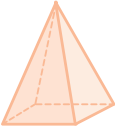
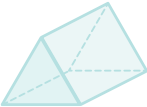
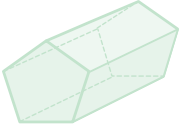
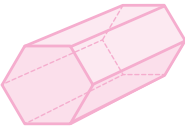
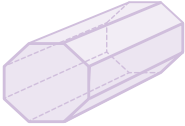
3D Shape Nets

Challenge

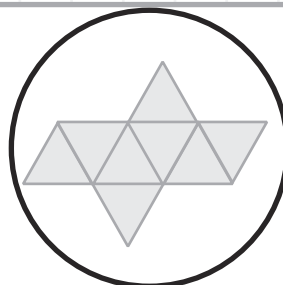
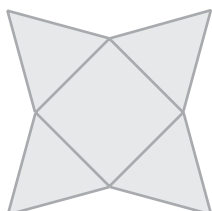
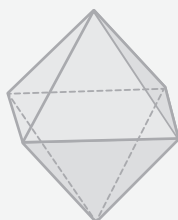
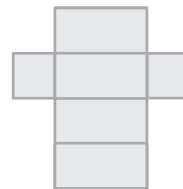
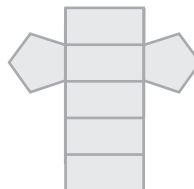
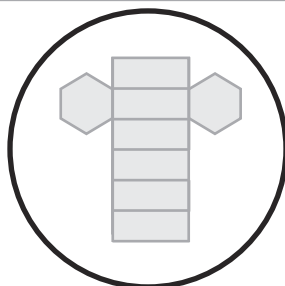
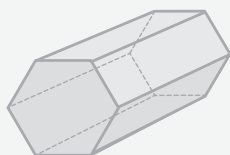
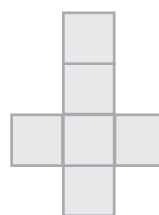
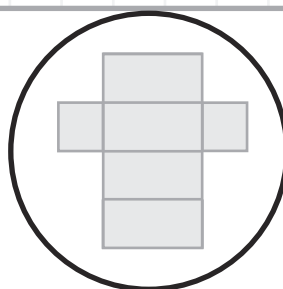
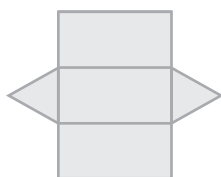
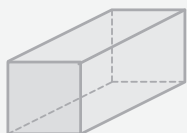
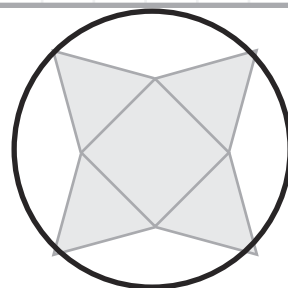
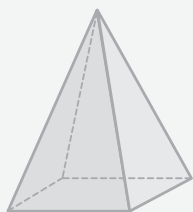
Which is the correct net for this cube?



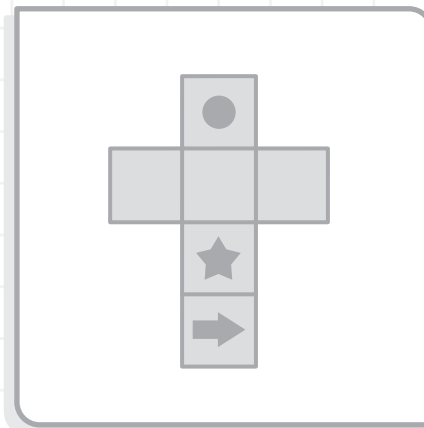
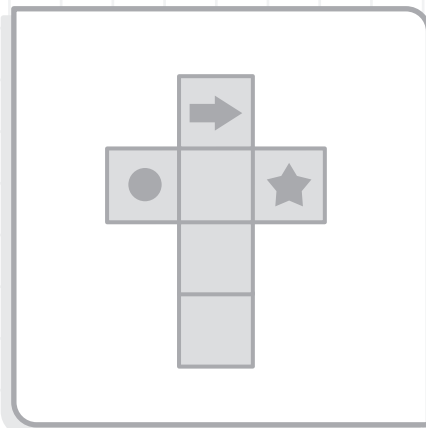
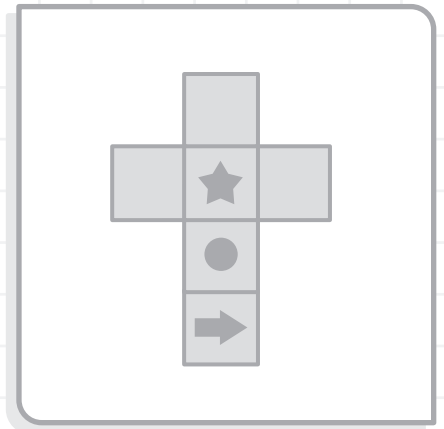
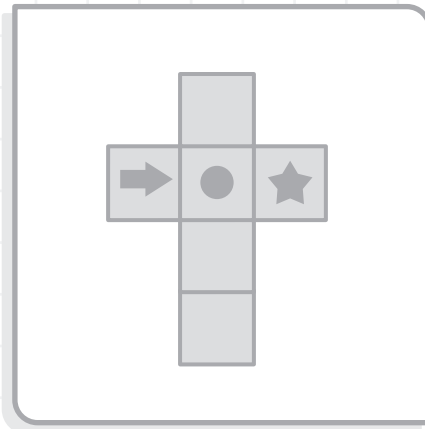
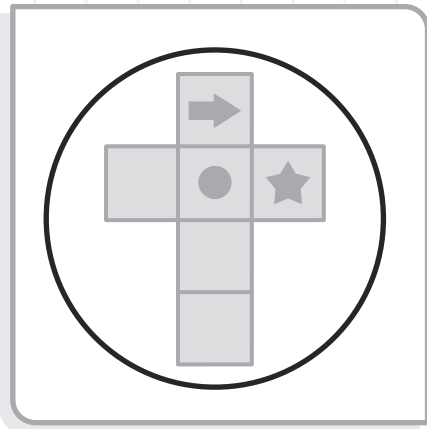
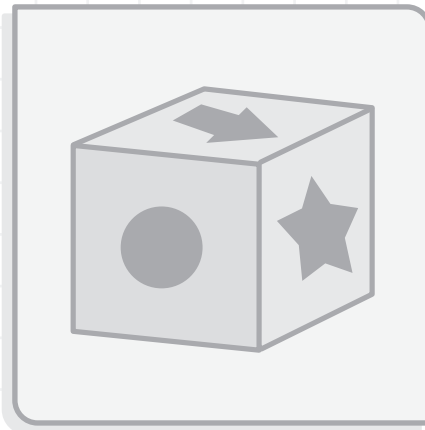
Properties of 3D Shapes

	Shape Name	Number of Faces	Types of Faces	Number of Edges	Number of Vertices	Possible Net
	cube	6	6 squares	12	8	
						
						
						
						
						
						
						
						

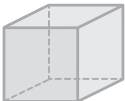
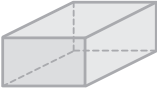
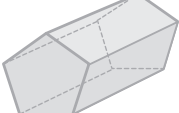
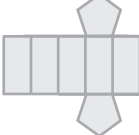
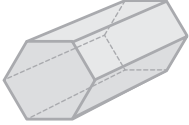
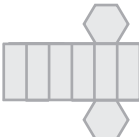
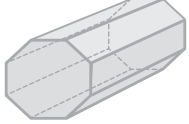
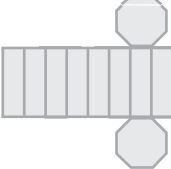
3D Shape Nets - Answers



3D Shape Nets - Answers



Properties of 3D Shapes – Answers

	Shape Name	Number of Faces	Types of Faces	Number of Edges	Number of Vertices	Possible Net
	cube	6	6 squares	12	8	
	cuboid	6	6 rectangles	12	8	
	tetrahedron	4	4 equilateral triangles	6	4	
	square-based pyramid	5	1 square 4 triangles	8	5	
	triangular prism	5	2 triangles 3 rectangles	9	6	
	pentagonal prism	7	2 pentagons 5 rectangles	15	10	
	hexagonal prism	8	2 hexagons 6 rectangles	18	12	
	octahedron	8	8 equilateral triangles	12	6	
	octagonal prism	10	2 octagons 8 rectangles	24	16	

We hope you find the information on our website and resources useful. As far as possible, the contents of this resource are reflective of current professional research. However, please be aware that every child is different and information can quickly become out of date. The information given here is intended for general guidance purposes only and may not apply to your specific situation.



visit [twinkl.com](https://www.twinkl.com)

